

Sharpness-Aware Minimization with Adaptive Regularization for Training Deep Neural Networks

Jinping Zou | Xiaoge Deng | Tao Sun

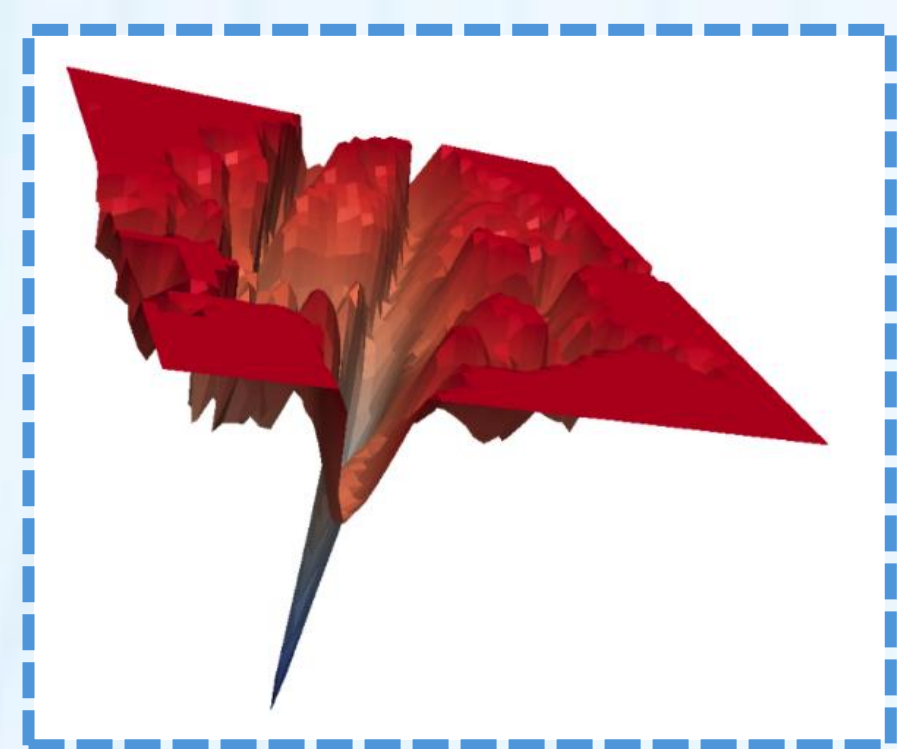
College of Computer Science and Technology

National University of Defense Technology, Changsha, China

{zoujinping, dengxg}@nudt.edu.cn, suntao.saltfish@outlook.com

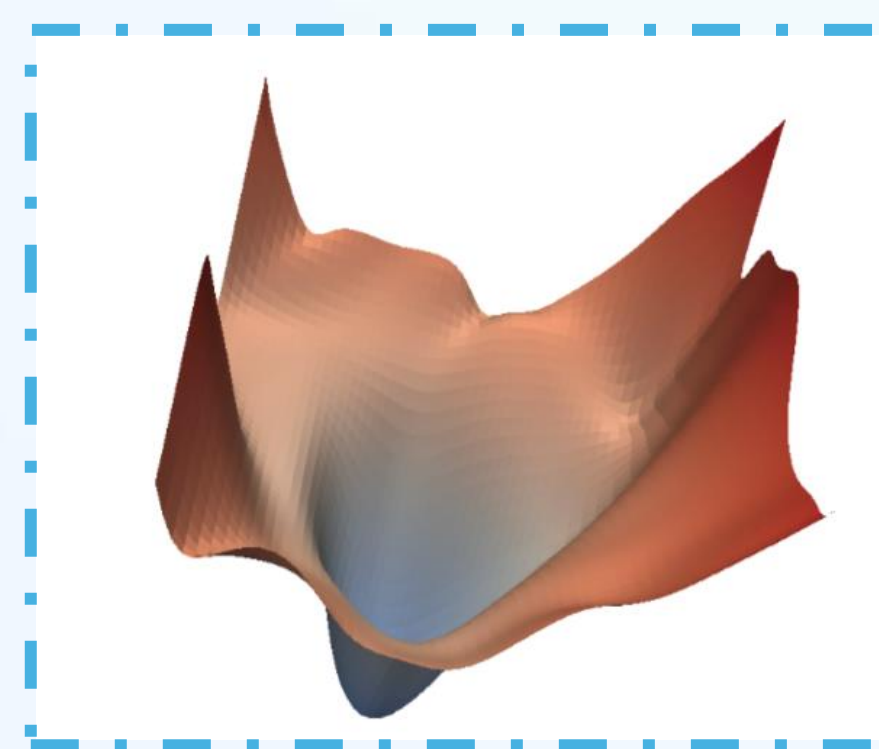
1. Background: Sharpness-Aware Minimization*

- Sharpness-Aware Minimization (SAM) is a novel numerical optimization algorithm that enhances generalization by achieving a flatter loss landscape.



Training with SGD*

Sharp minima
VS
Flat minima
↓
Flat minima can generalize better.



Training with SAM*

- Although SAM has proven effective in improving generalization, its objective function employs a fixed weight ratio between the loss term and sharpness term. Given that different deep learning tasks exhibit varying sensitivity to sharpness, the fixed weight ratio may limit SAM's broader applicability.

In this work, we propose an **adaptive weight scheme** for the sharpness term, which **flexibly modulates the regularization effect** in accordance with the sharpness ratio of the optimization process.

2. Deep Comparison and Motivation

- Difference of Optimization Objective Between SGD and SAM

(Stochastic Gradient Descent) SGD	(Sharpness-Aware Minimization) SAM
To solve $\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$,	To solve $\min_{\mathbf{x} \in \mathbb{R}^d} \max_{\ \epsilon\ \leq \rho} f(\mathbf{x} + \epsilon)$,
Construct iterable sequence:	Construct iterable sequence:
$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$.	$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k + \epsilon_k)$, $\epsilon_k = \rho \frac{\nabla f(\mathbf{x}_k)}{\ \nabla f(\mathbf{x}_k)\ }$.
Only minimize training loss.	Minimize the maximal loss within neighborhood.
$-\nabla f(\mathbf{x}_k)$: the <i>steepest descent direction</i> .	$-\nabla f(\mathbf{x}_k + \epsilon_k)$: the <i>direction generalizes better</i> .

- Rethink the Objective Function of SAM

Simple rewriting	loss	sharpness
$\min_{\mathbf{x} \in \mathbb{R}^d} \max_{\ \epsilon\ \leq \rho} f(\mathbf{x} + \epsilon) = \min_{\mathbf{x} \in \mathbb{R}^d} \left[1 \cdot f(\mathbf{x}) + 1 \cdot \max_{\ \epsilon\ \leq \rho} [f(\mathbf{x} + \epsilon) - f(\mathbf{x})] \right]$.		
The weight ratio is fixed between loss term and sharpness term (1:1), which gives rise to the following issues:		
- Limited flexibility - Deficient dynamics		

5. Conclusions

- We propose SAMAR algorithm to overcome the fixed weight ratio shortage.
- Theoretically, SAMAR converges at rate of $\mathcal{O}(1/\sqrt{K})$.
- Image classification task showcases SAMAR gains preferable improvement.

Future Work

- Test SAMAR on more complex **ImageNet** dataset.
- Extend to other settings liking **machine translation**, **transfer learning**, etc.

3. Method

Introduction of Adaptive Weight

Introducing adaptive weight into the objective function:

$$\min_{\mathbf{x} \in \mathbb{R}^d} \left[1 \cdot f(\mathbf{x}) + \lambda_k \cdot \max_{\|\epsilon\| \leq \rho} [f(\mathbf{x} + \epsilon) - f(\mathbf{x})] \right].$$

Construct numerical iteration:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta (\lambda_k \nabla f(\mathbf{x}_k + \epsilon_k) + (1 - \lambda_k) \nabla f(\mathbf{x}_k)),$$

$$\epsilon_k = \rho \frac{\nabla f(\mathbf{x}_k)}{\|\nabla f(\mathbf{x}_k)\|}.$$

Adaptive Regularization Strategy

Design the following *sharpness ratio rule*:

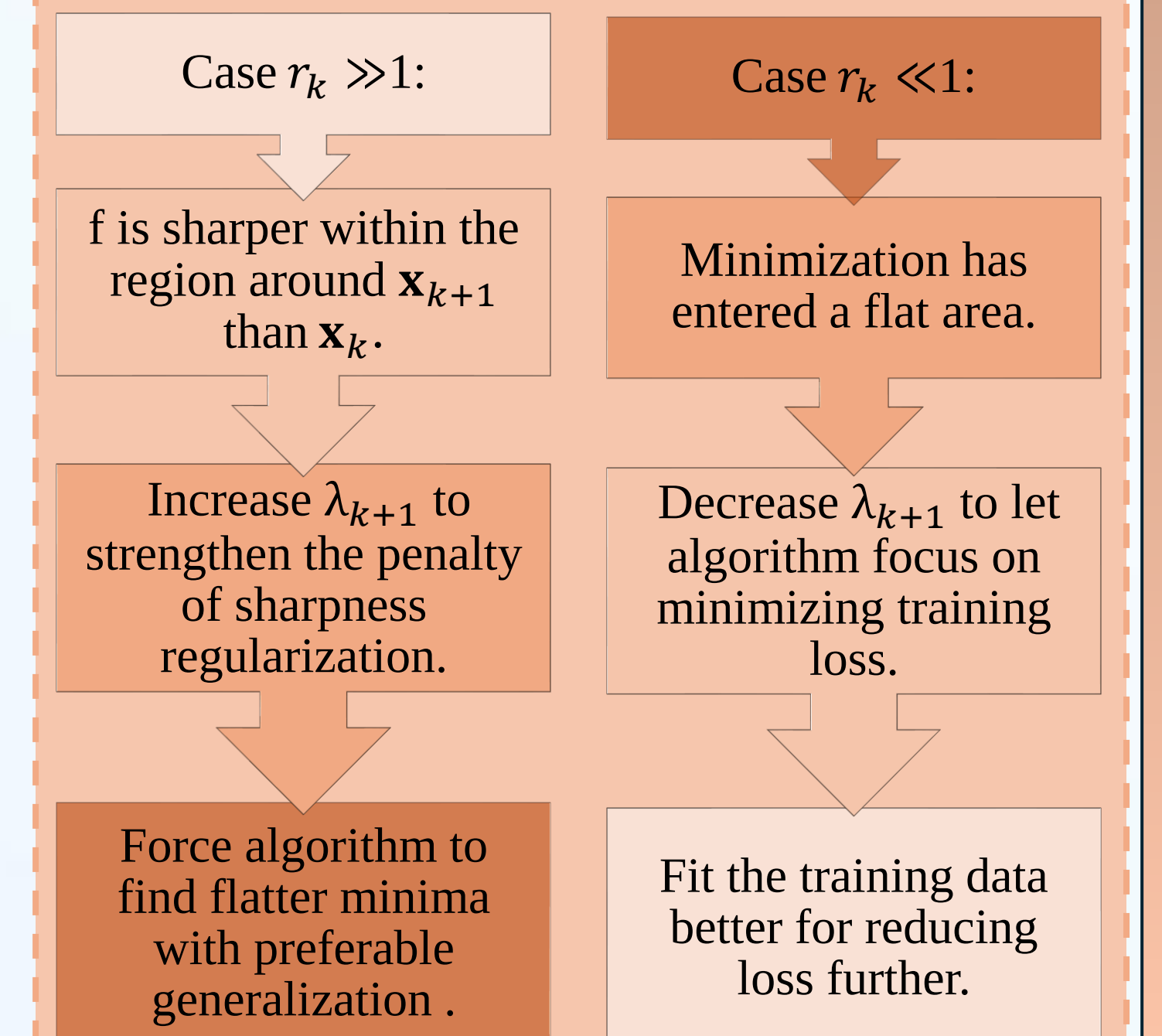
$$r_k \triangleq \frac{|f(\mathbf{x}_k + \epsilon_k) - f(\mathbf{x}_k)|}{|f(\mathbf{x}_{k-1} + \epsilon_{k-1}) - f(\mathbf{x}_{k-1})|}$$

$$\approx \frac{|\langle \nabla f(\mathbf{x}_k), \epsilon_k \rangle|}{|\langle \nabla f(\mathbf{x}_{k-1}), \epsilon_{k-1} \rangle|}$$

$$= \frac{\|\nabla f(\mathbf{x}_k)\|}{\|\nabla f(\mathbf{x}_{k-1})\|}$$

if $r_k \geq \chi$: $\lambda_{k+1} = \text{Proj}_{[\delta, 1-\delta]}(\gamma \lambda_k)$,
else: $\lambda_{k+1} = \text{Proj}_{[\delta, 1-\delta]}(\frac{1}{\gamma} \lambda_k)$,

Reasonable trade-off



4. Theoretical & Experimental

Theoretical Property

- The iterable sequence $\{\mathbf{x}_k\}$ generated by SAMAR algorithm converges at rate of $\mathcal{O}(1/\sqrt{K})$.

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{x}_k)\|^2] \leq \mathcal{O}(1/\sqrt{K}).$$

Algorithm 1 Sharpness Aware Minimization with Adaptive Regularization

Initialize: $\mathbf{x}_0, \rho, \eta, 0 < \lambda_0 < 1, \chi, \gamma > 1, 0 < \delta < 1$.

for $k = 0, \dots, K-1$ **do**

step 1: Compute the stochastic gradient $\mathbf{g}(\mathbf{x}_k)$ on the batch data.

step 2: Calculate $\epsilon_{\mathbf{x}_k} = \rho \frac{\mathbf{g}(\mathbf{x}_k)}{\|\mathbf{g}(\mathbf{x}_k)\|}$ through first-order stochastic linearization approximation.

step 3: Let $\mathbf{s}_k = (1 - \lambda_k) \mathbf{g}(\mathbf{x}_k) + \lambda_k \mathbf{g}(\mathbf{x}_k + \epsilon_{\mathbf{x}_k})$.

step 4: $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \mathbf{s}_k$.

step 5: Calculate $r_k = \frac{\|\mathbf{g}(\mathbf{x}_k)\|}{\|\mathbf{g}(\mathbf{x}_{k-1})\|}$.

step 6: if $r_k \geq \chi$ or $k = 0$: $\lambda_{k+1} = \text{Proj}_{[\delta, 1-\delta]}(\gamma \lambda_k)$.
 else: $\lambda_{k+1} = \text{Proj}_{[\delta, 1-\delta]}(\frac{1}{\gamma} \lambda_k)$.

step 7: $k = k + 1$.

end for

return $\mathbf{x}_K$

Experimental Result

- We train ResNet-34-10 neural network on CIFAR-100 dataset using SAMAR/SGD/SAM/VaSSO optimizers and record the top-1 accuracy. Training curvatures conform that SAMAR outperforms other rivals.



Reference

*Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *9th International Conference on Learning Representations, ICLR, 2021*.